AI, Top Earners, and Inequality in Transition Economies

How Machine Learning Concentrates Income at the Top of Middle-Income Distributions

This note formalizes a sharper inequality channel than the Gini-level test: AI raises inequality in transition (middle-income) economies primarily by lifting the **top income share**. The mechanism rests on AI being both *skill-complementary* and *capital-embodied*, so its gains accrue to a thin elite of high-skill workers and capital owners — a group that is especially small and concentrated in transition economies. Each block gives the equation and the meaning of its parameters.

1. Income structure and the top share

Partition workers into high-skill H (the top earners — managers, professionals in knowledge-intensive services) and low/mid-skill L. Output is produced with three factors: AI-embodied capital K, high-skill labor H, and low/mid-skill labor L. Total income is

$$Y_{it} = w_{it}^{H} H_{it} + w_{it}^{L} L_{it} + R_{it} K_{it}, \tag{1}$$

and the top income share — the empirical object of interest (e.g. top 1% or top 10%) — is

$$S_{it}^{P} = \frac{w_{it}^{H} H_{it} + \kappa_{it}\, R_{it} K_{it}}{Y_{it}}. \tag{2}$$

- w^{H}, w^{L}, R — prices of high-skill labor, low/mid-skill labor, and capital.
- $\kappa_{it} \in [0,1]$ — fraction of capital income accruing to top earners (ownership concentration), typically *high* in transition economies.
- S^{P} — top earners' share of total income; rises with the skill premium and with capital's share.

2. Production: AI as skill-complementary, capital-embodied technology

A three-factor CES with $\rho = \frac{\sigma-1}{\sigma}$:

$$Y_{it} = \Big[\beta_K \big(A_t^{K}(G_t)\, K_{it}\big)^{\rho} + \beta_H \big(A_t^{H}(G_t)\, H_{it}\big)^{\rho} + \beta_L \big(A_t^{L}(G_t)\, L_{it}\big)^{\rho}\Big]^{1/\rho}. \tag{3}$$

- $\sigma > 1$ — elasticity of substitution; factors are gross substitutes (the empirically relevant case for AI capital vs. routine labor).
- $A_t^{K}(G_t)$ — AI-augmentation of capital, $A^{K\prime} > 0$ (AI is embodied in capital).
- $A_t^{H}(G_t)$ — AI-augmentation of high-skill labor, $A^{H\prime} > 0$ (AI complements cognitive skill).
- $A_t^{L}(G_t)$ — AI effect on low/mid-skill labor, $A^{L\prime} \leq 0$ (routine displacement).
- $\beta_K, \beta_H, \beta_L$ — factor weights.

3. Factor prices and the skill premium

Marginal products give factor prices (with $\Sigma_{it} = \sum_j \beta_j (A_j X_j)^\rho$):

$$w_{it}^H = \beta_H \big(A^H\big)^\rho H_{it}^{\rho-1} Y_{it}^{1-\rho}, \qquad R_{it} = \beta_K \big(A^K\big)^\rho K_{it}^{\rho-1} Y_{it}^{1-\rho}. \tag{4}$$

The skill premium is

$$\frac{w_{it}^H}{w_{it}^L} = \frac{\beta_H}{\beta_L} \Big(\frac{A^H}{A^L}\Big)^\rho \Big(\frac{H_{it}}{L_{it}}\Big)^{\rho-1}. \tag{5}$$

- Since AI raises A^H/A^L and $\rho > 0$ (when $\sigma > 1$), the skill premium *rises* with G.
- The premium also rises as high-skill labor becomes *scarcer* (H/L small) — the case in transition economies with thin tertiary-educated workforces.

4. Factor shares and the top share

The income share of factor j is

$$s_{it}^j = \frac{\beta_j \big(A_t^j X_{j,it}\big)^\rho}{\Sigma_{it}}, \qquad j \in \{K, H, L\}, \quad \sum_j s^j = 1, \tag{6}$$

so the top share is the sum of the high-skill labor share and the elite-captured part of capital's share:

$$S_{it}^P = s_{it}^H + \kappa_{it}\, s_{it}^K. \tag{7}$$

5. Main result: AI raises the top share

The log-derivative of any factor share with respect to AI is

$$\frac{\partial \ln s_{it}^j}{\partial G_t} = \rho \Big(\frac{A^{j\prime}}{A^j} - \sum_k s_{it}^k \frac{A^{k\prime}}{A^k} \Big). \tag{8}$$

A factor gains income share iff its AI-augmentation rate exceeds the share-weighted average augmentation rate. Because $A^{K\prime}/A^K$ and $A^{H\prime}/A^H$ both exceed the average, while $A^{L\prime}/A^L \le 0$ drags it down, both top-earner components gain at the expense of s^L. The effect on the top share is

$$\boxed{\frac{\partial S_{it}^P}{\partial G_t} = \rho \Big[s^H \Big(\tfrac{A^{H\prime}}{A^H} - \bar{a} \Big) + \kappa\, s^K \Big(\tfrac{A^{K\prime}}{A^K} - \bar{a} \Big) \Big] > 0, \qquad \bar{a} \equiv \sum_k s^k \tfrac{A^{k\prime}}{A^k}.} \tag{9}$$

- $\bar{a}$ — share-weighted average augmentation rate (the "hurdle" a factor must clear to gain share).
- The displaced share comes out of s^L — the bottom and middle of the distribution.
- Result: AI is *top-share-increasing* whenever it augments capital and high skill faster than the average factor, the empirically standard configuration.

6. Why the effect is larger in transition economies

Let $\mu_{it} \in (0,1)$ be the fraction of the workforce with enough human capital to complement AI (absorptive capacity), so effective high-skill labor is $H_{it} = \mu_{it}\bar{H}_{it}$. The *concentration* of top income — income per top earner relative to the mean — is

$$C_{it}^{P} \equiv \frac{S_{it}^{P}}{\mu_{it}}. \tag{10}$$

Differentiating,

$$\frac{\partial C_{it}^{P}}{\partial G_t} = \frac{1}{\mu_{it}} \frac{\partial S_{it}^{P}}{\partial G_t} \; > \; \frac{\partial S_{it}^{P}}{\partial G_t} \quad \text{whenever } \mu_{it} < 1. \tag{11}$$

- μ_{it} small (few workers can use AI) and κ_{it} large (concentrated capital ownership) are precisely the conditions of transition economies.
- Same AI shock ⇒ gains are split among fewer people and more of capital income flows to the elite ⇒ a *sharper* rise in measured top concentration than in advanced economies, where broad human capital spreads AI's gains. The amplifier is $1/\mu_{it}$.

7. The FinTech ambiguity

FinTech enters the top share with two opposing forces:

$$\frac{\partial S_{it}^{P}}{\partial F_t} = \underbrace{-\,\xi_{\text{inc}}}_{\substack{\text{inclusion: bottom gains credit}}} + \underbrace{\xi_{\text{elite}}\, G_t}_{\substack{\text{elite capture via algorithmic finance}}}. \tag{12}$$

- $\xi_{\text{inc}} > 0$ — financial-inclusion effect: FinTech extends credit to the bottom, lowering S^P.
- $\xi_{\text{elite}} > 0$ — elite-capture effect: where FinTech mainly arms the digital elite, it *raises* S^P, and this term *scales with* G (algorithmic finance compounds AI's concentration).
- Net sign is ambiguous: equalizing if inclusion dominates, dis-equalizing if elite capture does — consistent with weak/insignificant cross-country Gini results, which average over both forces.

8. Sectoral link: polarization, not just inequality

Connecting to the structural-transformation model, top income gains arise from KIS expansion (high w^H) while displacement hits routine manufacturing/clerical work:

$$\Delta S_{it}^{P} \;\approx\; \underbrace{\tfrac{w^H}{Y}\, p_k\, \Delta\theta_{ikt}}_{\substack{\text{top gains from KIS growth}}} - \underbrace{\tfrac{w^L}{Y}\, p_m\, \Delta\theta_{imt}}_{\substack{\text{middle hollowed by displacement}}}. \tag{13}$$

The middle of the distribution thins (routine jobs lost) while the top expands (KIS jobs gained) — *job polarization*, with the top share rising even if the very bottom is partly cushioned by informal-sector absorption.

9. Empirical specification

With top income shares as the outcome (World Inequality Database, top 1% / top 10% pre-tax):

$$\text{Top}_{it} = a + b_1 G_{it-1} + b_2\left(\frac{G_{it-1}}{H_{it-1}}\right) + b_3 F_{it-1} + b_4\big(F_{it-1}G_{it-1}\big) + b_5\big(G_{it-1}{\cdot}MI_{it}\big) + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{14}$$

- $b_1 > 0$ — AI raises the top share (the baseline result of Section 5).
- $b_2 > 0$ — the effect is stronger where human capital is scarce (the $1/\mu$ amplifier of Section 6).
- $b_5 > 0$ — the effect is stronger in transition/middle-income economies (the headline claim).
- b_4 — ambiguous: positive if FinTech amplifies elite capture, negative if inclusion offsets it.

Mediation (does concentration feed back to growth?). First stage AI → top share, second stage top share → growth:

$$\text{Top}_{it} = \gamma_0 + \gamma_1 G_{it-1} + \Lambda' X_{it-1} + \mu_i + \lambda_t + u_{it}, \tag{15}$$

$$g_{it} = \delta_0 + \delta_1 G_{it-1} + \theta\,\widehat{\text{Top}}_{it-1} + \Psi' X_{it-1} + \mu_i + \lambda_t + e_{it}. \tag{16}$$

A negative θ would say AI-driven concentration is itself a drag on growth — the inequality channel through which AI could deepen, rather than relieve, the middle-income trap.

Data. WID (wid.world) for top income shares; SWIID for Gini robustness; Barro–Lee / PWT for $\bar{H}$; World Bank classifications for MI_{it}.